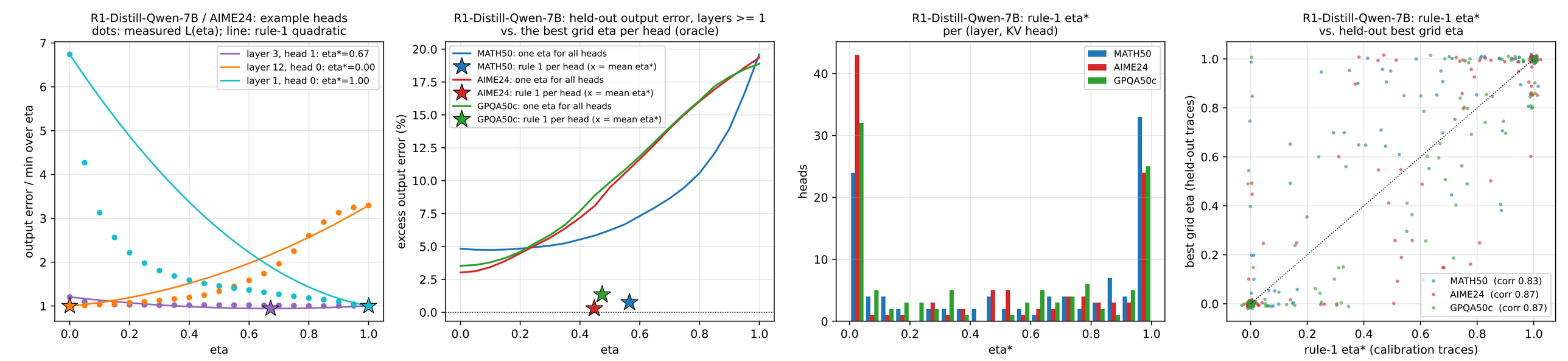

R1-Distill-Qwen-7B / AIME24: example heads
dots: measured L(eta); line: rule-1 quadratic
layer 3, head 1: eta*=0.67
layer 12, head 0: eta*=0.00
layer 1, head 0: eta*=1.00
output error / min over eta
eta
R1-Distill-Qwen-7B: held-out output error, layers >= 1
vs. the best grid eta per head (oracle)
MATH50: one eta for all heads
MATH50: rule 1 per head (x = mean eta*)
AIME24: one eta for all heads
AIME24: rule 1 per head (x = mean eta*)
GPQA50c: one eta for all heads
GPQA50c: rule 1 per head (x = mean eta*)
excess output error (%)
eta
R1-Distill-Qwen-7B: rule-1 eta*
per (layer, KV head)
MATH50
AIME24
GPQA50c
heads
eta*
R1-Distill-Qwen-7B: rule-1 eta*
vs. held-out best grid eta
MATH50 (corr 0.83)
AIME24 (corr 0.87)
GPQA50c (corr 0.87)
best grid eta (held-out traces)
rule-1 eta* (calibration traces)
R1-Distill-Llama-8B / AIME24: example heads
dots: measured L(eta); line: rule-1 quadratic
layer 5, head 6: eta*=0.47
layer 29, head 6: eta*=0.00
layer 8, head 4: eta*=1.00
output error / min over eta
eta
R1-Distill-Llama-8B: held-out output error, layers >= 1
vs. the best grid eta per head (oracle)
MATH50: one eta for all heads
MATH50: rule 1 per head (x = mean eta*)
AIME24: one eta for all heads
AIME24: rule 1 per head (x = mean eta*)
GPQA50c: one eta for all heads
GPQA50c: rule 1 per head (x = mean eta*)
excess output error (%)
eta
R1-Distill-Llama-8B: rule-1 eta*
per (layer, KV head)
MATH50
AIME24
GPQA50c
heads
eta*
R1-Distill-Llama-8B: rule-1 eta*
vs. held-out best grid eta
MATH50 (corr 0.45)
AIME24 (corr 0.54)
GPQA50c (corr 0.60)
best grid eta (held-out traces)
rule-1 eta* (calibration traces)